



United International University (UIU)

Dept. of Computer Science & Engineering (CSE)

Mid-Term Exam Trimester: Fall 2023

Course Code: CSE 4325 Course Title: Microprocessors and Microcontrollers

Total Marks: 30 Duration: 1 hour 45 minute(s)

Any examinee found adopting unfair means would be expelled from the trimester/ program as per UIU disciplinary rules.

Question 1: Answer all the questions. (6 Marks)

Transfer of bus control from processor to device takes **100 ns**. Transfer of bus control from device to processor takes **200 ns**.

a.	If one of the input/output devices employs DMA in burst mode and takes 10000300 ns to transfer 1536 bytes of data, what is the data transfer rate of the device in KB/s?	[2]
b.	Suppose, you are transferring 1536 bytes of data in both burst mode and cycle stealing mode. For the first one-third of the bytes, you use burst mode and for the rest two-third , you use cycle stealing mode . Assume that in cycle stealing mode, data is transferred 4 bytes at a time . How long will it take to transfer a block of 1536 bytes? (Use the data transfer rate found from (a))	[4]

Question 2: Answer all the questions. (6 Marks)

a.	Suppose, execution of a signed additional instruction (6BCDH + 5234H) occurred, What would be the value of zero flag (ZF), sign flag (SF), parity flag (PF), overflow flag (OF)?	[4]
b.	Describe in short the four general purpose registers of the 8086 microprocessor.	[2]

Question 3: Answer all the questions. (6 Marks)

a.	Suppose the address 79E4H : 2A8CH has an instruction. To access this instruction, what should be the value of I) the segment register if the offset register holds the value of 5A3CH. II) the segment register if the offset register holds the value of 5A2DH. III) the offset register if the segment register holds the value of 2679H. Justify whether your answer is valid or not.	[3]

b.	RAM[A] has a data bus of 32 bits and RAM[B] has an address bus of 26 bits. Both RAMs have a total memory capacity of 128 MB. What is the address bus width of RAM[A] and data bus width of RAM[B]?	[3]
<u>Question 4: Answer all the questions. (6 Marks)</u>		
a.	Draw the diagram of a microprocessor with a 20-bit address bus and 8-bit data bus interfaced to 28 KB RAM system using the full decoding method . Each RAM chip has a 13-bit address bus and 8-bit data bus. Provide the corresponding address range (starting address and end address) for the system.	[3]
b.	Modify the circuit of (a) to address memory range B0000H – B6FFFH. Draw the diagram as before and provide the modified address range. You need only draw the part of the diagram that goes through any modification.	[3]

<u>Question 5: Answer all the questions. (6 Marks)</u>		
a.	Calculate the memory location of CS and IP for the interrupt INT EH in the interrupt vector table.	[3]
b.	What is the sequence of events that takes place when a software interrupt instruction is executed?	[2]
c.	Explain the function of “IRET” in interrupt service procedure.	[1]